Scaled Forecast Error Metrics

A practical guide to MASE, RMSSE and MSSE — with retail and supply-chain examples

The central question

A forecast error of 100 units might be excellent for one product and poor for another. Scaled metrics put errors into context by comparing a model's error with the error from a simple benchmark forecast.

Quick takeaway

Keep WAPE for overall business accuracy. Add MASE or RMSSE when you need a benchmark-relative view across products with different volumes and demand patterns. A scaled metric complements WAPE; it does not automatically replace it.

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Prepared as a non-interactive reference document. Examples are illustrative and are not actual competition or company results.

1. Why do we need a scaled metric?

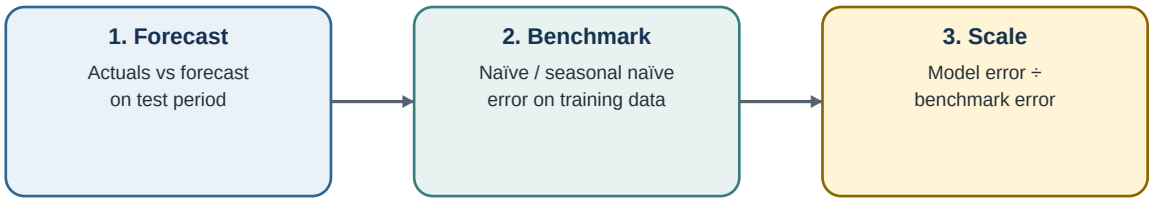
Forecast errors are measured in the units of the target: units sold, units returned, cases, or revenue. A raw error is hard to compare across products because products have different scales and different levels of natural variation.

Example	Product A: bestseller	Product B: slow mover
Typical weekly demand	1,000 units	10 units
Your model's absolute error	100 units	5 units
Simple benchmark's historical MAE	200 units	2 units
Model error ÷ benchmark error	100 ÷ 200 = 0.50	5 ÷ 2 = 2.50

Interpretation

Product A has the larger error in units, but its error is half the benchmark's historical error. Product B has a smaller error in units, but its error is 2.5 times the benchmark's historical error. The scaled comparison adds context that raw units alone do not provide.

The flow of a scaled metric



Interpretation

Below 1: lower error than the historical benchmark error

Above 1: higher error than the historical benchmark error

Important precision: the denominator is not generic volatility. It is the error of a defined naïve or seasonal-naïve benchmark, calculated on the training data. The numerator is the model's error on the test period.

A score below 1 means the model's test error is below the benchmark's training error under the chosen scaling. It does not prove that the model beat the benchmark on the same test period; compare both models on the same test window for that claim.

2. The main scaled error metrics

Let y be actual demand, ■ be forecast demand, T the number of training observations, H the number of test observations, and m the benchmark lag. Use m = 1 for a non-seasonal naïve benchmark, or a seasonal lag such as 52 for weekly data with annual seasonality.

A. MASE — Mean Absolute Scaled Error

MASE = (test-set MAE of the model) ÷ (training-set MAE of the naïve benchmark)

MASE uses absolute errors. It is relatively less sensitive to very large individual misses than squared-error metrics. It is unit-free and lower is better.

B. MSSE — Mean Squared Scaled Error

MSSE = (test-set MSE of the model) ÷ (training-set MSE of the naïve benchmark)

MSSE squares errors before averaging, so large misses receive disproportionate influence. It is dimensionless, but its scale is squared; RMSSE is often easier to communicate.

C. RMSSE — Root Mean Squared Scaled Error

RMSSE = √MSSE

RMSSE takes the square root of the scaled squared-error ratio. It retains the stronger penalty for large errors while returning to the original error scale relative to the benchmark.

Metric	Numerator	Denominator	Main interpretation
MASE	Mean absolute test error	Mean absolute benchmark error on training data	Relative typical absolute error
MSSE	Mean squared test error	Mean squared benchmark error on training data	Relative squared-error performance
RMSSE	Square root of MSSE	Implicitly scaled by benchmark RMSE	Relative RMSE-style performance

How to read a score

MASE = 0.5 means the model's test MAE is half the benchmark's training MAE. RMSSE = 1.2 means the model's test RMSE is 1.2 times the benchmark's training RMSE. Neither metric is a percentage error.

3. Worked example: calculate MASE, MSSE and RMSSE

Suppose a product has this short training series. We use a non-seasonal naïve benchmark, so each historical value is compared with the previous observation.

Training week	Actual demand	Absolute change from prior week	Squared change
1	100	—	—
2	110	10	100
3	90	20	400
4	100	10	100
Average benchmark error	—	13.33	200

Benchmark denominators

Mean absolute benchmark error = (10 + 20 + 10) ÷ 3 = 13.33

Mean squared benchmark error = (100 + 400 + 100) ÷ 3 = 200

Now evaluate two test weeks:

Test week	Actual	Forecast	Absolute error	Squared error
5	120	110	10	100
6	100	110	10	100
Average	—	—	10	100

Final calculations

MASE = 10 ÷ 13.33 ≈ 0.75

MSSE = 100 ÷ 200 = 0.50

RMSSE = √0.50 ≈ 0.707

What this says

Relative to the historical naïve benchmark error, the model's absolute test error is lower (MASE below 1), and its squared test error is also lower (RMSSE below 1). For a fair model-versus-baseline test, also forecast the same test weeks with the naïve model and compare both on those weeks.

4. WAPE versus scaled metrics

WAPE and scaled metrics answer different questions. WAPE is usually intuitive for business reporting; scaled metrics add a benchmark-relative view of performance across series.

WAPE (often called weighted MAPE in some teams)

$$\text{WAPE} = \Sigma|\text{actual} - \text{forecast}| \div \Sigma|\text{actual}|$$

When actual demand is non-negative, WAPE expresses total absolute error as a share of total actual volume. The aggregation naturally gives high-volume products more influence.

Question	WAPE	MASE / RMSSE
What is the denominator?	Total absolute actuals in the evaluation window	Historical naïve/seasonal-naïve error on training data
Do high-volume products dominate?	Usually, yes, in pooled WAPE	Not if you calculate per product and then take a simple mean
Can zero-demand days be included?	Yes; their absolute errors enter the numerator, but they add nothing to the denominator	Yes; errors are included in the numerator; benchmark differences determine the scale
Can the metric be undefined?	Yes, if total absolute actuals are zero	Yes, if the training benchmark error is zero
Best for	Overall volume-weighted accuracy	Relative comparison across heterogeneous series
Main caution	Can hide poor performance on low-volume products	Aggregation choices and benchmark choice affect interpretation

Example: why WAPE can be dominated by a bestseller

Product	Actual volume	Absolute error	Product-level error / actual
Bestseller	10,000	1,000	10%
Slow mover	100	50	50%
Pooled total	10,100	1,050	WAPE = 10.4%

The pooled WAPE is about 10.4%, even though the slow mover has a 50% error relative to its own volume. MASE/RMSSE can help assess that product relative to its own benchmark, but they do not automatically measure its business importance.

5. When should you use a scaled metric?

Situation	Scaled metric useful?	Why / caution
Many SKUs with very different sales volumes	Yes	Adds a per-series benchmark-relative view alongside WAPE.
Assortment and ranging analysis	Often	Helps compare forecast skill across products, but not a substitute for commercial potential or margin.
Intermittent demand / spare parts	Potentially	Does not divide by each test actual; however, benchmark denominator can still be zero.
Model selection (e.g., Seasonal Naïve vs AutoARIMA vs LightGBM)	Yes	Compare each model on the same test windows; scaled scores complement direct benchmark comparison.
Long-range planning (e.g., 60-week returns forecast)	Yes, as a complement	Report by horizon bucket; long horizon alone does not make scaled metrics preferable.
One aggregate business forecast	Usually alongside WAPE	WAPE is easier to explain in volume terms; calculate aggregate-series MASE separately if useful.
Financial or operational decisions	Not by itself	Also evaluate cost, stockouts, capacity, service level, margin or other business outcomes.
Series with constant values / zero benchmark error	Not directly	MASE/RMSSE are undefined; define a documented policy or use another metric.

When not to use scaled metrics as the only score

Do not use MASE/RMSSE alone when the business needs total volume accuracy, when benchmark errors are zero or unstable, or when a forecast's business cost is asymmetric. For example, underforecasting returns may create capacity problems, while overforecasting may lead to excess staffing. A symmetric error metric does not encode those costs.

Rule of thumb

Use WAPE to answer “How wrong are we relative to the total volume?” Use MASE/RMSSE to answer “How good is the model relative to a simple benchmark for each series?” Use business metrics to answer “What does this error cost us?”

6. Aggregation, seasonal benchmarks and common pitfalls

A. Product-level versus aggregate-level MASE/RMSSE

Method	How it works	Interpretation
Mean of product-level scores	Calculate MASE/RMSSE for each product, then average scores	Typical relative performance across products; simple mean gives each valid product equal weight.
Score on aggregate series	Sum products into a department total, then calculate the metric on that time series	Relative accuracy of the department-level total. Errors can cancel when signed forecasts are aggregated before absolute error is calculated.

These are different metrics and can lead to different conclusions. Report which aggregation method you use. For product-level summaries, consider median MASE/RMSSE and the percentage of products beating a benchmark, because a few extreme scores can distort the mean.

B. Pick a benchmark that makes sense

For non-seasonal data, a common scale is the one-step naïve forecast: compare each training observation with the previous one. For weekly data with annual seasonality, a seasonal naïve benchmark may compare a week with the corresponding week 52 weeks earlier. Use a seasonal lag only when supported by the data and the forecast design.

C. Zero denominators and intermittent series

Having some non-zero demand does not guarantee a valid scaled metric. The denominator is based on differences at the chosen lag. If all such differences are zero, the denominator is zero. For intermittent demand, document how these series are handled rather than silently dropping them.

D. Rolling-origin evaluation and leakage

At each backtest origin, calculate the benchmark scale using only the training observations available at that origin. Evaluate the model and benchmark on the same held-out horizon. Recalculate the scale as the training window changes if that is your chosen protocol.

E. RMSSE's sensitivity to large errors

Because errors are squared, a few large misses can have a strong influence on RMSSE. This can be useful when big misses matter, but it can also make the score less representative of typical errors. Review error distributions and business impact alongside it.

7. Applying this to a 60-week Return Planning Forecast

For a long-range returns forecast, do not switch from WAPE to a scaled metric simply because the horizon is 60 weeks. Use metrics together because they describe different aspects of performance.

Suggested evaluation design

Evaluation layer	Recommended reporting
Overall return volume	WAPE and bias
Different product / category series	Product-level MASE or RMSSE, summarised with median, mean and % beating benchmark
Forecast horizon	Report metrics for weeks 1–13, 14–26, 27–39 and 40–60
Aggregate planning level	WAPE on the aggregated total; optionally calculate separate MASE/RMSSE on the aggregate series
Operational outcomes	Warehouse capacity, staffing, service level, excess capacity and costs where measurable
Baseline comparison	Evaluate the chosen naïve / seasonal-naïve benchmark on the same held-out periods as the candidate model

A note on forecasting returns

Returns can be influenced by sales volumes, product mix, return rates, seasonality, promotions and changes in policy or customer behaviour. If possible, assess both return volume and return rate, and distinguish forecast errors caused by a changed sales plan from errors in the return-rate model.

Suggested reporting dashboard

Metric	Overall	Weeks 1–13	Weeks 14–26	Weeks 27–39	Weeks 40–60
WAPE	✓	✓	✓	✓	✓
Bias	✓	✓	✓	✓	✓
Median MASE/RMSSE	✓	✓	✓	✓	✓
% series beating baseline	✓	✓	✓	✓	✓
Business impact	✓	As relevant	As relevant	As relevant	As relevant

Recommended starting point

Keep WAPE as the headline volume-accuracy metric. Add RMSSE or MASE as a diagnostic for cross-series relative performance. Track bias, report by horizon bucket, and evaluate operational impact. Confirm benchmark choice and zero-denominator handling before adopting scaled scores in production.

8. Implementation checklist and glossary

Before you publish MASE/RMSSE results

■ Define the series grain: SKU, SKU-location, category, or aggregate.

■ Define the benchmark: naïve or seasonal naïve, with the correct lag.

■ Calculate the scaling denominator using training data only at each forecast origin.

■ Compare model and benchmark on the same test observations.

■ Document zero-denominator and missing-data handling.

■ Report the aggregation method: mean of per-series scores or score on an aggregate series.

■ Report results by forecast horizon, not only one pooled number.

■ Pair scaled scores with WAPE, bias and business outcome measures.

Glossary

Term	Meaning
Actual (y)	Observed demand or returns.
Forecast (■)	Model's predicted demand or returns.
Naïve forecast	Forecast the next value using the most recent actual value.
Seasonal naïve	Forecast using the value from the corresponding season in the previous cycle.
MAE	Mean absolute error: average of |actual – forecast|.
MSE	Mean squared error: average of (actual – forecast)².
WAPE	Sum of absolute errors divided by sum of absolute actuals.
MASE	Test MAE divided by training naïve/seasonal-naïve MAE.
MSSE	Test MSE divided by training naïve/seasonal-naïve MSE.
RMSSE	Square root of MSSE.
Rolling-origin backtest	Repeatedly train at past cutoffs and evaluate forecasts on later, unseen periods.

Further reading

• Rob J. Hyndman & Anne B. Koehler (2006), “Another look at measures of forecast accuracy,” International Journal of Forecasting. DOI: 10.1016/j.ijforecast.2006.03.001

• M5 forecasting competition resources and evaluation discussions for RMSSE in hierarchical retail demand forecasting.

• Forecast evaluation best practice: use time-aware backtesting and select metrics aligned with the decisions the forecast supports.

Final takeaway: no single metric tells the whole story. WAPE explains total volume error; MASE/RMSSE provide benchmark-relative insight across series; bias and business metrics explain operational consequences.